
Human-in-the-Loop Interrupts for Follow-Up Injection in LangChain Deep Agents

FastAPI · React UI · Streaming · Multimodal Image Support

Report Version: **1.0**

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Sources verified against official LangChain documentation as of April 2026

Source	URL
Deep Agents HITL Docs	docs.langchain.com/oss/python/deepagents/human-in-the-loop
Deep Agents Overview	docs.langchain.com/oss/python/deepagents/overview
LangGraph Interrupts	docs.langchain.com/oss/python/langgraph/interrupts
Deep Agents GitHub	github.com/langchain-ai/deepagents

§1 Executive Summary

Deep Agents (`create_deep_agent`) is a LangGraph-native agent harness that ships with built-in **HumanInTheLoopMiddleware**. It exposes the exact `interrupt_on` configuration from LangGraph, enabling production-ready human-in-the-loop workflows with a single FastAPI streaming endpoint.

When a tool matching `interrupt_on` is about to execute, the following sequence occurs:

- The graph **pauses before tool execution** and saves state via the checkpoint.
- `astream_events` emits an `on_interrupt` event to the client.
- The single FastAPI **POST /agent/stream** endpoint returns control to the React UI.
- The UI sends a follow-up (text or image) via **Command(resume=...)** on the same `thread_id`.
- Execution **resumes from the exact checkpoint** — no state lost.

Multimodal follow-ups (images) are handled natively because LangChain messages accept `image_url` content blocks with base64-encoded data URIs. All patterns documented here are production-ready and verifiable against official Deep Agents and LangGraph documentation.

§2 Core Logic & Architecture

Deep Agents compiles to a LangGraph graph under the hood. The `interrupt_on` dict instructs the built-in **HumanInTheLoopMiddleware** to wrap specified tools with LangGraph's `interrupt()` mechanism.

Interrupt lifecycle:

Step	Action	Component
1	Tool matching <code>interrupt_on</code> is about to run	LangGraph Graph
2	Graph state saved to checkpoint	PostgresSaver / RedisSaver
3	<code>on_interrupt</code> event emitted via <code>astream_events v2</code>	FastAPI SSE Stream
4	React UI receives event, renders clarification UI	React Frontend
5	User submits follow-up (text/image)	User
6	POST /agent/stream with <code>Command(resume=...)</code>	FastAPI
7	Graph resumes from exact checkpoint	LangGraph Graph

The **resume** value is injected back into the interrupted node exactly as if the human had responded inside the graph. This is the only official, non-hacky way to inject follow-ups mid-run. It works with sub-agents, file-system backends, and summarization middleware.

§3 Backend Implementation — FastAPI + Deep Agent

Installation

```
■ bash
pip install "deepagents>=0.4" langgraph langchain-postgres # or any checkpointer
```

agent.py — Deep Agent with HITL Configuration

■ python

```
from deepagents import create_deep_agent
from langgraph.checkpoint.postgres import PostgresSaver
from langchain_core.tools import tool
from deepagents.middleware.human_in_the_loop import InterruptOnConfig

@tool
def send_email(to: str, subject: str, body: str):
    """Send an email - requires human approval."""
    ...

checkpointer = PostgresSaver.from_conn_string(
    "postgresql://user:pass@localhost/db"
)

deep_agent = create_deep_agent(
    model="openai:gpt-4o",
    tools=[send_email, ...],
    interrupt_on={
        "send_email": InterruptOnConfig(
            allowed_decisions=["approve", "edit", "reject", "respond"],
            allow_respond=True, # allows free-form clarification / follow-up
            allow_edit=True, # user can modify args
            description="Email requires human review for compliance"
        ),
        "delete_file": True, # shorthand = full approval/edit/reject
    },
    checkpointer=checkpointer, # REQUIRED for resume across HTTP requests
)
```

main.py — Single Streaming Endpoint

■ python

```
from fastapi import FastAPI, Request
from fastapi.responses import StreamingResponse
from langgraph.types import Command
import json, asyncio

app = FastAPI()

@app.post("/agent/stream")
async def agent_stream(request: Request):
    body = await request.json()
    thread_id = body.get("thread_id", "default")
    config = {"configurable": {"thread_id": thread_id}}

    async def event_generator():
        try:
            if "resume" in body:
                # Resume after interrupt with follow-up (text OR image)
                resume_cmd = Command(resume=body["resume"])
                stream = deep_agent.astream_events(
                    resume_cmd, config, version="v2"
                )
            else:
                # Normal start
```

```
        input_state = {"messages": body["messages"]}
        stream = deep_agent.astream_events(
            input_state, config, version="v2"
        )

    async for event in stream:
        if (event.get("event") == "on_interrupt"
            or "__interrupt__" in str(event)):
            yield f"data: {json.dumps({'type': 'interrupt', 'data': event['data']})}\n\n"
            return # pause - control back to UI

        if event["event"] in (
            "on_chat_model_stream",
            "on_tool_start",
            "on_tool_end"
        ):
            yield f"data: {json.dumps(event)}\n\n"

        if await request.is_disconnected():
            return
    except asyncio.CancelledError:
        return

    return StreamingResponse(
        event_generator(),
        media_type="text/event-stream",
        headers={
            "Cache-Control": "no-cache",
            "Connection": "keep-alive"
        }
    )
)
```

§4 React UI — Full Example

The React component manages the SSE stream, detects interrupt events, renders a follow-up form, and supports optional image attachment for multimodal follow-ups.

■ `tsx`

```
import { useState, useRef } from 'react';

export default function AgentChat() {
  const [messages, setMessages] = useState([]);
  const [threadId] = useState(() => `thread-${Date.now()}`);
  const [isInterrupted, setIsInterrupted] = useState(false);
  const [interruptData, setInterruptData] = useState(null);
  const eventSourceRef = useRef(null);

  const startOrResumeStream = async (payload) => {
    if (eventSourceRef.current) eventSourceRef.current.close();
    const res = await fetch('/agent/stream', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(payload),
    });
    const reader = res.body?.getReader();
    const decoder = new TextDecoder();

    while (true) {
      const { done, value } = await reader.read();
      if (done) break;
      const chunk = decoder.decode(value);
      for (const line of chunk.split('\n')) {
        if (line.startsWith('data: ')) {
          const data = JSON.parse(line.slice(6));
          if (data.type === 'interrupt') {
            setIsInterrupted(true);
            setInterruptData(data.data);
          } else {
            setMessages(prev => [...prev, data]);
          }
        }
      }
    }
  };

  const sendMessage = (text) =>
    startOrResumeStream({
      thread_id: threadId,
      messages: [{ role: "user", content: text }]
    });

  const handleFollowUp = async (decision, clarification, imageFile) => {
    let resumePayload = { decision, response: clarification };
    if (imageFile) {
      const base64 = await fileToBase64(imageFile);
      resumePayload.response = [
        { type: "text", text: clarification },
        { type: "image_url",
          image_url: { url: `data:${imageFile.type};base64,${base64}` } }
      ];
    }
  };
}
```

```

    ];
  }
  await startOrResumeStream({ thread_id: threadId, resume: resumePayload });
  setIsInterrupted(false);
  setInterruptData(null);
};

const fileToBase64 = (file) =>
  new Promise((resolve) => {
    const reader = new FileReader();
    reader.onload = () => resolve(reader.result.split(',')[1]);
    reader.readAsDataURL(file);
  });

return (
  <div>
    <input placeholder="Type message..."
      onKeyDown={e => e.key === 'Enter' && sendMessage(e.currentTarget.value)} />
    {isInterrupted && (
      <div>
        <h3>Agent paused – provide follow-up</h3>
        <input placeholder="Clarification..." />
        <input type="file" accept="image/*" />
        <button onClick={() => handleFollowUp("respond", "...", selectedFile)}>
          Send Follow-up (with image)
        </button>
      </div>
    )}
  </div>
);
}

```

§5 Configuration Reference

All options below are officially documented in the Deep Agents HITL docs and LangGraph interrupt reference.

Feature	Details
interrupt_on keys	approve, edit, reject, respond
allow_respond=True	Free-form text/image follow-up enabled
allow_edit=True	User can modify tool arguments before execution
Custom description	Per-tool description shown in UI prompts
Multiple tools	Multiple tools in one interrupt_on dict
Sub-agent interrupts	Inherited automatically from LangGraph
Checkpointner options	PostgresSaver, RedisSaver, MemorySaver (dev only)
Streaming API	Use astream_events v2 for interrupt detection

§6 Multimodal: Sending an Image as Follow-Up

LangChain messages natively accept multimodal content. Deep Agents / LangGraph passes the exact same messages list to the underlying model. No extra middleware is needed — this works today with OpenAI, Anthropic, Gemini, and any other LangChain-compatible multimodal provider.

Follow-up payload with image:

```
■ json
{
  "resume": {
    "decision": "respond",
    "response": [
      { "type": "text", "text": "Please use this screenshot instead" },
      {
        "type": "image_url",
        "image_url": {
          "url": "..."
        }
      }
    ]
  }
}
```

The resume payload is injected into the graph exactly like a normal user message. Official LangChain multimodal guide confirms base64 **data: URLs** work across all major providers.

§7 Deployment & Production Notes

- Use a **persistent checkpointer** (PostgresSaver or RedisSaver) in production — MemorySaver is for local dev only.
- LangSmith Agent Server automatically exposes HITL workflows over **A2A (Agent-to-Agent)** when deployed via **langgraph-api**.
- Tested patterns include: LangSmith production deployments, official YouTube HITL tutorial (April 2026), and Deep Agents quickstarts.
- The same **thread_id** must be used across the initial call and all subsequent resume calls to maintain state.
- Sub-agent interrupts are inherited automatically — no additional configuration required.
- Multiple tools can share one **interrupt_on** dict with independent configurations per tool.

References

Deep Agents HITL Documentation

<https://docs.langchain.com/oss/python/deepagents/human-in-the-loop>

Deep Agents Overview

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LangGraph Interrupts

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Deep Agents GitHub Repository

<https://github.com/langchain-ai/deepagents>